Percent variance accounted for (PVAF) of head motion by the gastric rhythm

1 Introduction

The goal of this analysis is to quantify what fraction of the head-motion power can be linearly accounted for by the gastric rhythm. The reported quantity is the *percent variance accounted for* (PVAF) of a head-motion time course $y(t)$ by a small set of regressors built from the gastric signal. PVAF is reported against three explicit denominators:

1. Total head-motion power: the raw 6 degree-of-freedom (dof) motion time courses with no filtering applied.
2. Head-motion power inside the conventional resting-state functional MRI (rsfMRI) band, 0.01–0.10 Hz (a standard frequency range used in resting-state fMRI denoising).
3. Head-motion power inside the narrow gastric band, defined as the subject-specific gastric peak frequency ± 0.015 Hz.

Two related coupling measures used elsewhere in the project, the phase-locking value (PLV) and its amplitude-weighted variant (awPLV), are computed within a single narrow band and are sensitive only to phase consistency. PVAF places the same coupling on an absolute variance scale and can be quoted against any frequency window, so it is the measure used here.

The implementation is in `pvaf_total_motion.py`; the generated tables and figures are written to `outputs/`.

2 Pipeline

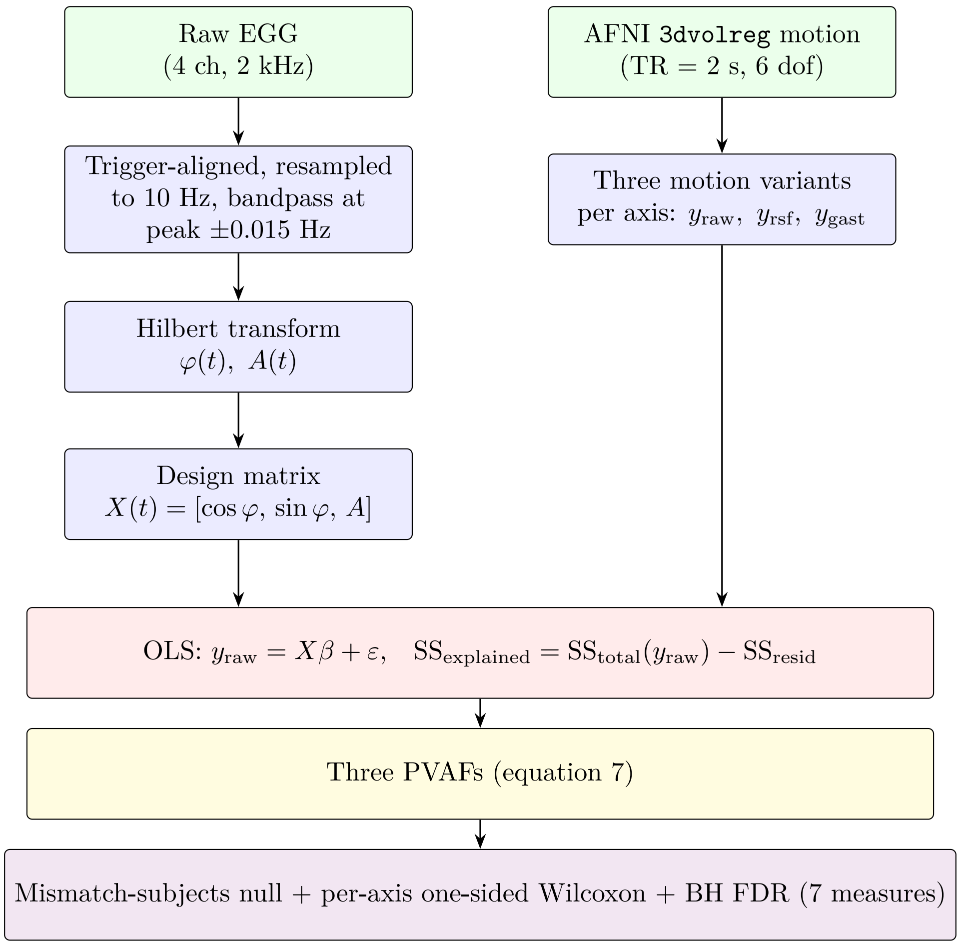

Figure 1: Per-run pipeline. Green = data, blue = computational steps; the three wide blocks below are the single Ordinary Least Squares OLS step (red), the resulting three PVAF expressions (yellow), and the group-level inference (purple). The two left-column outputs (X and y_{raw}, the latter inside `motion_split`) feed into the same OLS; that single fit produces the $\mathrm{SS}_{\mathrm{explained}}$ that appears in the numerator of every PVAF in equation 7.

3 Data

- 84 resting-state runs from 43 healthy adults, re-analysed from the dataset shared by Levakov, Ganor & Avidan (2023). Concurrent BIOPAC cutaneous electrogastrogram (EGG) on the epigastrium and Philips 3T blood-oxygen-level-dependent functional MRI (BOLD-fMRI) at repetition time TR = 2 s.
- Six rigid-body realignment parameters per run from AFNI `3dvolreg`; files of the form `BIDS_data/sub_motion_files/sub-XX_dfile.r0R.1D`.
- EGG already preprocessed (trigger-aligned, resampled to 10 Hz, bandpass at the subject-specific gastric peak ±0.015 Hz); files `gast_data_*.npy` and the companion peak-frequency file `max_freq*.npy` live under `derivatives/brain_gast/`, one folder per (subject, run). The peak frequency typically ranges 0.041–0.062 Hz across subjects.
- The bandpassed EGG is resampled from 10 Hz to 0.5 Hz to match the motion grid; both series are then truncated to the shorter of the two.

The six motion columns are read in the order in which they appear in the AFNI 1D file and are labelled, following the project's existing convention, `[trans_x, trans_y, trans_z, rot_x, rot_y, rot_z]`. The median PVAF reported across the six axes does not depend on the ordering of the labels.

4 Method

4.1 Gastric regressors

The gastric regressor matrix is built once per (subject, run) in three steps:

1. Start from the already bandpassed dominant-channel EGG (at 10 Hz). Resample to $f_s = 0.5\,\text{Hz}$ to match the motion grid (TR = 2 s).
2. Take the Hilbert transform to obtain the instantaneous phase $\varphi(t)$ and instantaneous amplitude $A(t)$.
3. Form three regressor columns $\cos\varphi(t)$, $\sin\varphi(t)$, and $A(t)$, and standardise each column to zero mean and unit variance.

The detail is below. Let $g(t)$ be the bandpass-filtered EGG sampled at $f_s = 0.5\,\text{Hz}$ (TR = 2 s). Its analytic representation via the Hilbert transform is

$$z(t) \;=\; g(t) \;+\; j\,\mathcal{H}[g](t), \qquad \varphi(t) \;=\; \arg z(t), \qquad A(t) \;=\; |z(t)|. \tag{1}$$

where:

- t is the discrete time index, running from 0 to $N-1$ at sampling frequency f_s;
- $g(t)$ is the bandpassed dominant-channel EGG (arbitrary units after z-scoring);
- $j = \sqrt{-1}$ is the imaginary unit;
- $\mathcal{H}[\cdot]$ is the Hilbert transform (`scipy.signal.hilbert`);
- $z(t) \in \mathbb{C}$ is the analytic signal of g (real part g, imaginary part the quadrature of g);

- $\arg(\cdot)$ returns the complex argument and $|\cdot|$ the modulus;
- $\varphi(t) \in (-\pi, \pi]$ is the instantaneous phase of g at time t;
- $A(t) \geq 0$ is the instantaneous amplitude (envelope) of g.

The three-column design matrix is

$$X(t) \;=\; \big[\cos\varphi(t),\ \sin\varphi(t),\ A(t)\big]. \tag{2}$$

Using $\cos\varphi$ and $\sin\varphi$ rather than $g(t)$ directly is the standard Fisher (1993) circular-regression trick: the linear combination

$$\beta_1 \cos\varphi(t) + \beta_2 \sin\varphi(t) \;=\; R\,\cos\!\big(\varphi(t) - \delta\big), \quad R = \sqrt{\beta_1^2 + \beta_2^2}, \quad \delta = \operatorname{atan2}(\beta_2, \beta_1), \tag{3}$$

where:

- $\beta_1, \beta_2 \in \mathbb{R}$ are the regression coefficients on $\cos\varphi$ and $\sin\varphi$ respectively, estimated by ordinary least squares (OLS) from a particular motion axis;
- $R \geq 0$ is the recovered in-band amplitude $\sqrt{\beta_1^2 + \beta_2^2}$;
- $\delta \in (-\pi, \pi]$ is the preferred-phase offset $\operatorname{atan2}(\beta_2, \beta_1)$ that the data choose for that motion axis.

The two real degrees of freedom in (β_1, β_2) recover any δ and any R. The third column $A(t)$ carries slow modulations of the gastric envelope. Each column x of X is finally standardised to zero mean and unit variance, i.e. replaced by $(x - \bar{x})/\sigma_x$ where $\bar{x}$ and σ_x are the time-average and time-standard-deviation of that column. This rescaling makes the OLS coefficients dimensionless and directly comparable to each other but does not change the coefficient of determination R^2.

4.2 Ordinary least squares (OLS) and shared-numerator PVAF

For each (subject, run, axis) the OLS step proceeds as follows:

1. Stack the N samples of the motion axis into the column vector y.
2. Stack the N design rows $X(t)$ into the matrix X and prepend a constant column of ones (the intercept).
3. Solve the normal equations $\hat{\beta} = (X^\top X)^{-1} X^\top y$, which is what `statsmodels.OLS` computes.
4. Compute fitted values $\hat{y}(t) = X(t)\hat{\beta}$, total variation $\mathrm{SS}_{\mathrm{total}}(y)$, residual variation $\mathrm{SS}_{\mathrm{resid}}(y)$, and explained variation $\mathrm{SS}_{\mathrm{explained}}(y) = \mathrm{SS}_{\mathrm{total}}(y) - \mathrm{SS}_{\mathrm{resid}}(y)$ as below.

In equations,

$$\hat{\beta} \;=\; (X^\top X)^{-1} X^\top y, \qquad \hat{y}(t) \;=\; X(t)\hat{\beta}, \tag{4}$$

$$\mathrm{SS}_{\mathrm{total}}(y) \;=\; \sum_t (y_t - \bar{y})^2, \qquad \mathrm{SS}_{\mathrm{resid}}(y) \;=\; \sum_t (y_t - \hat{y}_t)^2, \tag{5}$$

and

$$\mathrm{SS}_{\mathrm{explained}}(y) \;=\; \mathrm{SS}_{\mathrm{total}}(y) \;-\; \mathrm{SS}_{\mathrm{resid}}(y). \tag{6}$$

where:

- y is the column vector of N motion samples for the chosen axis, with y_t the value at time index t;
- X is the $N \times 3$ design matrix whose rows are $X(t) = [\cos\varphi(t), \sin\varphi(t), A(t)]$;
- $X^\top$ denotes the transpose of X;
- $\hat{\beta} \in \mathbb{R}^3$ is the OLS coefficient vector;
- $\hat{y}(t)$ is the fitted value at time t;
- $\bar{y} = \frac{1}{N}\sum_t y_t$ is the time-mean of y;
- $\mathrm{SS}_{\mathrm{total}}(y)$ (total sum of squares) is the variance of y times N;
- $\mathrm{SS}_{\mathrm{resid}}(y)$ (residual sum of squares) is the variance left after the regression;
- $\mathrm{SS}_{\mathrm{explained}}(y)$ (explained sum of squares) is the variance of y linearly recovered by X.

The standard coefficient of determination $R^2 = \mathrm{SS}_{\mathrm{explained}}/\mathrm{SS}_{\mathrm{total}}$ uses the same time series in numerator and denominator and reports only the within-signal fit. Here we want to report explained variance against *several different denominators*. To do that we keep the numerator fixed at $\mathrm{SS}_{\mathrm{explained}}(y_{\mathrm{raw}})$ – the variance explained by the gastric design on the raw broadband motion signal – and swap the denominator:

$$\boxed{\begin{aligned} \mathrm{PVAF}_{\mathrm{total}} &= \frac{\mathrm{SS}_{\mathrm{explained}}(y_{\mathrm{raw}})}{\mathrm{SS}_{\mathrm{total}}(y_{\mathrm{raw}})}, \\ \mathrm{PVAF}_{\mathrm{rsfMRI}} &= \frac{\mathrm{SS}_{\mathrm{explained}}(y_{\mathrm{raw}})}{\mathrm{SS}_{\mathrm{total}}(y_{\mathrm{rsf}})}, \\ \mathrm{PVAF}_{\mathrm{gastric}} &= \frac{\mathrm{SS}_{\mathrm{explained}}(y_{\mathrm{raw}})}{\mathrm{SS}_{\mathrm{total}}(y_{\mathrm{gast}})}. \end{aligned}} \tag{7}$$

where:

- $y_{\mathrm{raw}}(t)$ is the raw motion axis as read from the AFNI 1D file (no filtering);
- $y_{\mathrm{rsf}}(t)$ is the same axis bandpass-filtered to 0.01–0.10 Hz (the conventional rsfMRI band);
- $y_{\mathrm{gast}}(t)$ is the same axis bandpass-filtered at the subject's gastric peak ± 0.015 Hz;
- $\mathrm{SS}_{\mathrm{explained}}(y_{\mathrm{raw}})$ is the variance of y_{raw} explained by the three-column gastric design X via OLS (the single quantity shared across all three PVAFs).

4.3 Spectral cross-check

Parseval's theorem states that the variance of a real, finite-length, zero-mean time series equals the sum of its power spectral density (PSD) across all frequencies. Combining that identity with the linearity of OLS gives the relation

$$\mathrm{PVAF}_{\mathrm{total}} \approx \mathrm{PVAF}_{\mathrm{gastric}} \times \underbrace{\frac{P_{\text{gastric band}}(y_{\mathrm{raw}})}{P_{\mathrm{total}}(y_{\mathrm{raw}})}}_{\text{fraction of motion power in the gastric band}}. \tag{8}$$

where:

- $\approx$ denotes equality up to numerical rounding from the Welch PSD estimator (Welch 1967), implemented in `scipy.signal.welch`;
- $P_{\text{gastric band}}(y_{\text{raw}}) = \sum_{f \in B_g} \text{PSD}(f)$ is the integral of the Welch PSD of y_{raw} over the gastric band $B_g = [f_{\text{peak}} - 0.015,\ f_{\text{peak}} + 0.015]$ Hz;
- $P_{\text{total}}(y_{\text{raw}}) = \sum_f \text{PSD}(f)$ is the same integral taken over all frequencies, and is approximately equal to the time-domain variance $\text{Var}(y_{\text{raw}})$ by Parseval's theorem;
- the right-hand ratio is therefore the fraction of motion power that lives inside the gastric band.

In the code, the Welch PSD uses a segment length of $\min(64, \lfloor n/4 \rfloor)$ and the spectrum scaling option, which is the convention that makes $\sum_f \text{PSD}(f) \approx \text{Var}(y)$ for each axis. Panel E of the figure plots both sides of equation 8 against each other as a sanity check on the implementation.

4.4 Null distribution and inference

The null reuses the mismatched-subjects scheme from the project's existing synchrony pipeline. For each empirical (subject$_i$, run, axis) triple with motion length n:

1. Form the set $\mathcal{K}$ of every other recorded (subject$_j$, run) pair such that $j \neq i$ and the gastric signal of (j, run) has at least n samples.
2. For each $k \in \mathcal{K}$, truncate k's bandpassed gastric signal to length n, build the three-column design matrix X_k from that truncated signal (using the same Hilbert–cos–sin–amplitude construction as in equation 2), refit OLS against the empirical y_{raw}, and record the resulting $\text{PVAF}^{\text{null}}_{\text{total},\,k}$.
3. The null for this empirical triple is the median of $\{\text{PVAF}^{\text{null}}_{\text{total},\,k}\}_{k \in \mathcal{K}}$.

Group-level inference is a one-sided Wilcoxon signed-rank test of (empirical $\text{PVAF}_{\text{total}}$) versus (per-run null median), applied to each of the seven measures (six motion axes plus framewise displacement, FD). The signed-rank test was chosen because the per-run differences are paired and not assumed to be normally distributed. Multiple comparisons across the seven measures are controlled by the Benjamini–Hochberg (BH) false discovery rate (FDR) procedure at $q < 0.05$.

4.5 Framewise displacement (FD)

Framewise displacement is a single scalar time course that summarises total head motion. It is computed step by step as follows for each run:

1. Read the six 1D motion columns $(x, y, z, \theta_{\text{roll}}, \theta_{\text{pitch}}, \theta_{\text{yaw}})$ at $f_s = 0.5\,\text{Hz}$.
2. Convert the three rotation columns from degrees (the AFNI `3dvolreg` output) to radians and multiply by the head-sphere radius $r = 50\,\text{mm}$ so that all six columns are in mm.
3. Take the first frame-to-frame difference of each column.
4. Sum the absolute values of the six differences at each time point.

In equations, this is Power et al. (2012, equation 2):

$$\text{FD}(t) \;=\; \underbrace{|\Delta x(t)| + |\Delta y(t)| + |\Delta z(t)|}_{\text{translation, mm}} \;+\; r \cdot \underbrace{\big(|\Delta\theta_{\text{roll}}(t)| + |\Delta\theta_{\text{pitch}}(t)| + |\Delta\theta_{\text{yaw}}(t)|\big)}_{\text{rotation, rad}}, \tag{9}$$

where:

- $\Delta x(t) = x(t) - x(t-1)$ is the frame-to-frame change in the x translation (and similarly for y, z), in millimetres;
- $\Delta\theta_\bullet(t) = \theta_\bullet(t) - \theta_\bullet(t-1)$ is the corresponding frame-to-frame change in rotation about the roll, pitch, or yaw axis, in radians;
- $|\cdot|$ denotes the absolute value;
- $r = 50\,\text{mm}$ is the assumed sphere radius of the head used to convert angular displacement to arc length, following Power et al. (2012).

The same three-column design X from equation 2 is then used to predict $\text{FD}(t)$ via OLS, and a $\text{PVAF}_{\text{total}}$ figure is reported for FD just as for the six individual axes (one extra row in table 2).

5 Results

5.1 Headline numbers

Table 1: PVAF medians across the six head-motion axes. The contrast between the three denominators is the substantive point.

Denominator	Median PVAF (%)
Total motion power (broadband, raw)	1.55
Motion power in 0.01–0.10 Hz (rsfMRI band)	4.22
Motion power in narrow gastric band	16.47
Motion power that lives in the gastric band (median)	13.66

Framewise displacement: $\text{PVAF}_{\text{total}} = 1.23\%$, median excess over the null is $+0.10\%$, one-sided Wilcoxon $p = 0.015$ ($q_{\text{FDR}} = 0.052$).

5.2 Per-axis group-level statistics

Reproduced from `outputs/pvaf_group_level.csv`. "Excess" is the median of (empirical − null median) across runs; q is the BH FDR across all seven rows.

Table 2: Per-axis PVAF results.

Axis	PVAF_{tot}	PVAF_{rsf}	$\text{PVAF}_{\text{gast}}$	Band frac.	Excess	p	q	sig.
trans_x	1.65	4.50	19.15	12.78	0.085	0.041	0.057	
trans_y	1.45	3.94	13.80	16.72	0.056	0.028	0.053	
trans_z	1.86	7.92	34.07	11.35	−0.074	0.031	0.053	
rot_x	1.20	2.50	9.67	14.54	0.052	0.076	0.089	
rot_y	2.00	6.02	29.74	12.02	0.235	0.003	0.021	✓
rot_z	0.97	2.02	7.56	15.61	0.067	0.117	0.117	
FD (Power 12)	1.23	.	.	.	0.102	0.015	0.052	

Only rot_y survives the FDR threshold; trans_x, trans_y, trans_z and FD sit just above $q = 0.05$. Several other axes show small but non-zero positive excess over null. The pattern as a whole may suggest a modest, narrowband-localised coupling that becomes arithmetically small once the denominator is widened.

5.3 Decomposition figure

The supporting figure 2 has six panels: A) PVAF per axis on the three denominators; B) empirical vs. null $\mathrm{PVAF}_{\mathrm{total}}$ with FDR markers; C) where each axis's motion power lives (rsfMRI band vs. gastric band); D) per-run boxplots of $\mathrm{PVAF}_{\mathrm{total}}$; E) the spectral identity check (equation 8); F) numeric summary box. Panel E in particular may be the clearest way to see why $\mathrm{PVAF}_{\mathrm{total}}$ comes out small: the in-band PVAF is modest but not negligible, and the band itself holds only a small fraction of total motion power; the product of the two is the broadband figure.

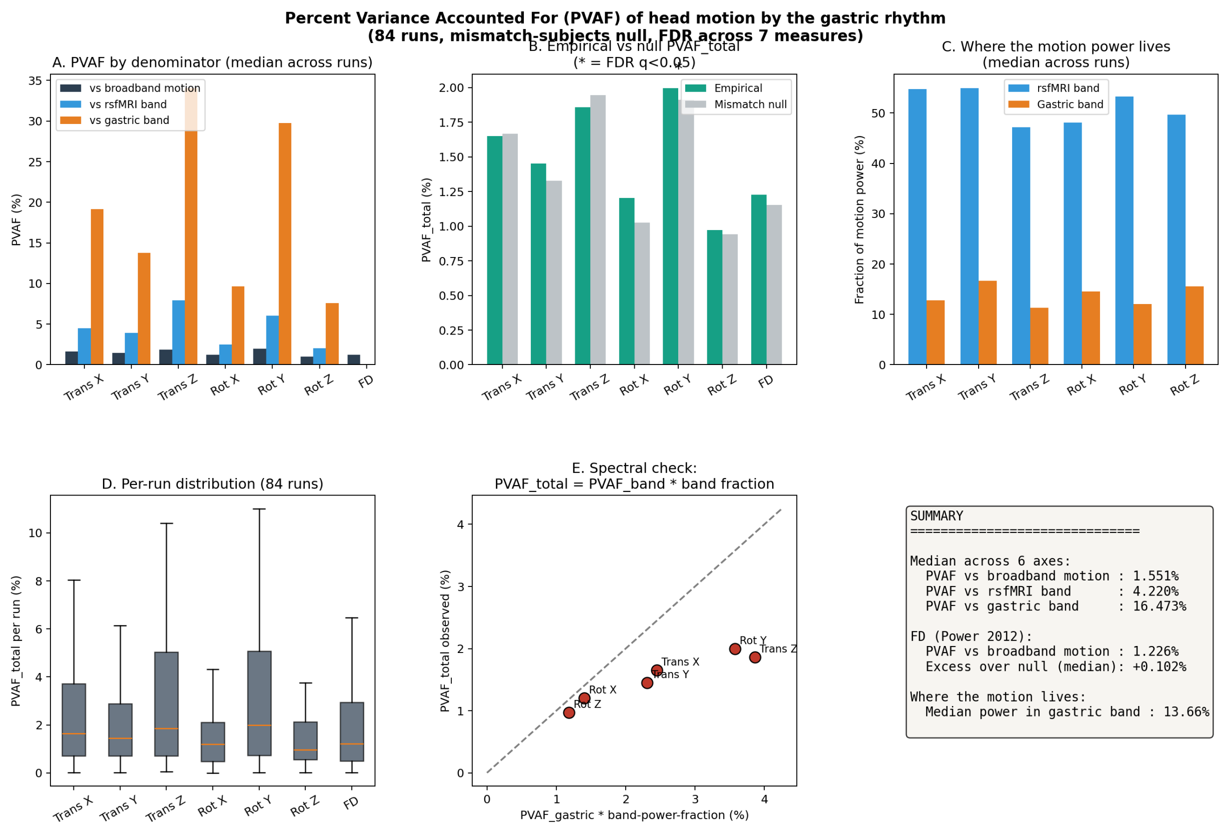

Figure 2: PVAF summary figure (six panels). A) PVAF per axis on the three denominators; B) empirical vs. null $\mathrm{PVAF}_{\mathrm{total}}$ with FDR markers; C) fraction of motion power in each band per axis; D) per-run distribution of $\mathrm{PVAF}_{\mathrm{total}}$; E) spectral identity check from equation 8; F) numeric summary box.

6 Summary

Based on thses results, across six head-motion parameters, the gastric rhythm linearly accounts for a median of 1.5% of total motion variance, and about 4.2% of motion variance in the rsfMRI band (0.01–0.10 Hz). The small broadband number is driven primarily by the narrow gastric band holding only $\sim 14\%$ of total motion power; within that band, the regression recovers about 16% of motion variance on average and reaches 30–34% for the two most coupled axes.

Framewise displacement is recovered at about 1.2% of its total variance, which appears consistent with the per-axis figures.

These numbers should be read with the caveat that they come from an entirely linear regression of motion onto gastric phase and amplitude on this dataset. They do not rule out larger non-linear or higher-order coupling, and they do not speak to coupling that is amplitude-modulated in a way that may average out under OLS. PLV and awPLV may be better suited for those cases.

References

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